Offline Digital Tools for Point Prevalence Surveys, Antimicrobial Consumption Monitoring and Ward-Level ADR Detection at Tanga Regional Referral Hospital

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Background: Hospital AMS teams in Tanzania are asked to do a lot with very little. They run point prevalence surveys, monitor antibiotic use and watch for adverse drug reactions, mostly on paper forms and spreadsheets that take days to clean and analyse. By the time a report reaches the committee, the data is often old. ADR reporting from the wards remains low, because busy ward staff do not see it as part of their daily work. We set out to build simple, affordable tools that let pharmacy staff collect, analyse and report their own data without internet access.

Methods: Drawing on three years of supporting the Tanga RRH AMS committee, we developed the Odieman suite, three linked offline tools. PPS Pro handles point prevalence surveys using WHO methodology. AMS Pro tracks antimicrobial consumption through DDD and DOT trends, Pareto analysis and cost estimates, and supports PDCA improvement cycles. VigiWard looks for ADR signals at ward level and analyses NCD and cardiometabolic patterns from routine pharmacy data. All three share one data source.

Results: The tools are in routine use by the AMS and pharmacovigilance teams at Tanga RRH. With PPS Pro, analysis is fully automated: once data collection ends, a complete PPS report is available with one click, while the findings are still relevant. Consumption and ADR data become committee-ready charts and reports the same way, on an ordinary computer without internet.

Conclusion: Practical digital tools can be built locally, by the people who do the work, at low cost. Timely PPS and consumption results allow early decisions on antibiotic use, while VigiWard brings pharmacovigilance closer to the wards, where most ADRs happen but few are reported. With support, these tools could reach other facilities and strengthen Tanzania's National Action Plan on AMR and ADR reporting to TMDA.

Keywords: antimicrobial stewardship, point prevalence survey, pharmacovigilance, digital health, Tanzania